
Department of Software Engineering

Bahria School of Engineering & Applied Sciences

Bahria University Islamabad Campus H-11

Assignment-4

CLO-3

(CSC 321) Design and Analysis of Algorithms

INSTRUCTIONS:

- A) Show all steps and reasoning.
 - B) Assignment to be handwritten.
 - C) Do your assignments on your notebook as advised in class. Then upload the PDF to BU-LMS.
 - D) Write on the top of the first page your Enrolment#, Name, Class, Course Title, Assignment#, and Date in order.
 - E) On the top-right corner of every subsequent page write your enrolment number and the page-number.
 - F) Start each question on a new page. Write prominently the question number.
 - G) Late assignments will not be accepted.
 - H) Viva after submission.
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You are tasked with analysing and solving a resource allocation problem in a multi-stage process. The problem involves managing a set of tasks with dependencies and constraints, using different algorithmic paradigms.

A company needs to allocate resources to complete a series of tasks efficiently.

The tasks are as follows:

Task	Start Time	End Time	Cost	Dependencies
T1	1	4	10	-
T2	3	5	15	T1
T3	0	6	20	-
T4	5	7	25	T2, T3
T5	8	9	30	T4
T6	5	9	50	T3

You need to perform the following tasks:

1. **Cost Optimization (Divide & Conquer):**

Develop a recursive approach to divide the tasks into smaller subproblems to find the subset of tasks with minimum total cost while ensuring dependencies are satisfied. Include a step-by-step explanation of your approach.

2. **Dependency Resolution (Decrease & Conquer):**

Explain how to resolve dependencies using a topological ordering of tasks. Show the step-by-step process to order tasks based on their dependencies and ensure that no task is executed before its prerequisite tasks are completed.

3. **Complexity Analysis:**

Analyse the time and space complexity for each paradigm used in your solution.
